

Evaluation Report

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The rapid advancement of Text-to-Speech (TTS) technologies has revolutionized the way we interact with machines, particularly in multilingual and dialect-rich regions like the Arab world. Arabic, with its complex linguistic structure and wide range of dialects—such as Modern Standard Arabic (MSA), Najdi, Hijazi, and Gulf—poses unique challenges for TTS systems in achieving both intelligibility and naturalness.

This evaluation report presents a comparative analysis of three state-of-the-art TTS models: Meta’s MMS (Massively Multilingual Speech), Coqui’s XTTS-v2, and Microsoft’s SpeechT5. Each model was tested using consistent text inputs across four dialects to assess their effectiveness in producing high-quality Arabic speech.

The evaluation is structured around both objective and subjective criteria. Objective metrics include latency (time taken to generate audio) and audio duration (length of the resulting speech). Subjective metrics—assessed through human listening—cover naturalness (how human-like the speech sounds) and clarity (ease of understanding).

1 MMS-TTS-ARA

The Massively Multilingual Speech (MMS) model developed by Meta AI aims to democratize speech technology by supporting over 1,000 languages. In this project, the MMS model was executed, specifically leveraging the pre-trained checkpoint facebook/mms-tts-ara, which is designed for Modern Standard Arabic (MSA). The evaluation, however, extended beyond MSA to include prominent regional dialects—Najdi, Hijazi, and Gulf—by inputting dialect-specific phrases directly into the MSA-trained model. While this approach allowed for broad testing, it also highlighted the model’s limitations in handling regional speech variations.

Objectively, the MMS model excelled in terms of latency and processing speed, achieving output generation times between 7.78 and 10.85 seconds, making it the fastest among the three models. The corresponding audio durations (7.23–9.49 seconds) aligned closely with natural speech pacing, suggesting good temporal alignment. From a computational standpoint, MMS is light-weight, fast to deploy, and suitable for real-time applications or low-resource environments.

However, subjective evaluation revealed notable drawbacks in terms of speech quality. The most significant issue was the robotic tone and lack of expressive prosody. Listeners consistently rated its naturalness at 2 out of 5, describing the speech as flat, mechanical, and emotionless. This limitation became more pronounced when the model was fed dialectal inputs; since MMS is not fine-tuned on these dialects, it produced outputs with a neutralized, sometimes awkward cadence. Despite these shortcomings, clarity remained strong (rated 4–5 out of 5), indicating the model’s phonetic accuracy and

intelligibility were intact even when dealing with varied linguistic inputs.

In terms of robustness, the MMS model handled short, affirmative phrases and moderately technical expressions related to automobiles with reasonable success. It did not produce errors or silence but rather delivered technically accurate yet emotionally disengaged output. However, it failed to adapt intonation or style to match regional characteristics or conversational tone, which are crucial for user engagement in natural human-computer interaction.

2 XTTS-v2 (Coqui TTS)

The XTTS-v2 model by Coqui.ai, implemented via `xttsv2.ipynb`, delivered the most natural and expressive Arabic speech among the evaluated models. Supporting cross-lingual synthesis and speaker embeddings, it performed exceptionally well across MSA, Najdi, Hijazi, and Gulf dialects. The speech output received top ratings for naturalness (5/5) and clarity (4–5/5), with strong handling of dialectal nuances, emotional tone, and pacing.

However, XTTS-v2 had the highest latency (45–64 seconds), making it less suitable for real-time applications. Despite this, it excelled in robustness, handling both short inputs and technical terms smoothly. Its ability to produce lifelike and dialect-sensitive speech makes it ideal for content creation, storytelling, and voice applications where quality outweighs speed.

3 SpeechT5 (Microsoft)

SpeechT5, implemented via the notebook `speecht5ttsclarttsar.ipynb`, offers a balanced blend of quality and performance. Trained for general-purpose TTS tasks, it produced clear and pleasant speech across MSA, Najdi, Hijazi, and Gulf dialects. It earned consistent scores of 4/5 for both naturalness and clarity, delivering smoother prosody than MMS and more efficient latency than XTTS-v2.

With latency ranging from 11.7 to 22.7 seconds, SpeechT5 is suitable for applications that need good-quality speech at moderate speed. It also handled both short responses and technical phrases well, maintaining intelligibility and flow. Overall, SpeechT5 is a reliable mid-tier option—ideal for interactive systems where quality and responsiveness must be balanced.

why speecht5

SpeechT5 emerged as a strong fit for this evaluation task, which required generating Arabic speech across dialects (MSA, Najdi, Hijazi, and Gulf) and testing robustness with short phrases and technical automotive terms, the model delivered a consistent balance of quality, adaptability, and performance. It produced speech that was rated 4 out of 5 in both naturalness and clarity, outperforming MMS in expressiveness while being significantly faster than XTTS-v2.

With moderate latency (11.7–22.7 seconds), SpeechT5 proved capable of handling diverse sentence structures without long wait times, making it well-suited for prototyping and iterative experimentation. It synthesized both short acknowledgments and longer technical sentences accurately, maintaining proper intonation and flow. While not fine-tuned on specific

Arabic dialects, its general-purpose architecture demonstrated strong generalization, effectively preserving dialectal flavor in output speech.

Given the task’s dual focus on performance metrics and linguistic variation, SpeechT5 offered an ideal middle ground: fast enough for multi-dialect testing, expressive enough to evaluate subjective quality, and robust enough to handle varied content. It’s particularly valuable when balancing accuracy, speed, and usability in multilingual TTS benchmarking.

4 Comparison Table

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Table 1: Comparison of TTS Models across Arabic Dialects

Model	Dialect	Latency (s)	Audio Duration (s)	Naturalness (1–5)	Clarity (1–5)
MMS	MSA	10.85	9.49	2	5
	Najdi	8.61	7.79	2	4
	Hijazi	9.52	7.81	2	4
	Gulf	7.78	7.23	2	4
XTTS-v2	MSA	46.22	6.87	5	5
	Najdi	64.16	9.24	5	4
	Hijazi	45.78	6.16	4	4
	Gulf	48.89	6.92	5	5
SpeechT5	MSA	22.73	7.33	4	4
	Najdi	14.21	7.01	4	4
	Hijazi	15.80	5.44	4	4
	Gulf	11.70	5.76	4	4

5 Conclusion

This evaluation set out to assess the performance of three advanced Text-to-Speech (TTS) models—Meta’s MMS, Coqui’s XTTS-v2, and Microsoft’s SpeechT5—across a range of Arabic dialects including Modern Standard Arabic (MSA), Najdi, Hijazi, and Gulf. The models were benchmarked using objective criteria such as latency and audio duration, as well as subjective criteria like naturalness, clarity, and robustness to varied input types.

While each model brought unique strengths to the table, SpeechT5 emerged as the most well-rounded and task-appropriate choice. It consistently delivered high-quality audio with strong ratings in both naturalness and clarity (4/5), maintained moderate latency suitable for interactive applications, and handled both short phrases and technical terminology with ease. Unlike MMS, which was limited to MSA and sounded robotic, and XTTS-v2, which offered high realism but at the cost of heavy latency, SpeechT5 offered a practical middle ground.

What further distinguishes SpeechT5 is that it is pre-trained not only on Modern Standard Arabic but also exposed

to multiple regional dialects, giving it a foundational advantage in handling linguistic variation. This makes it particularly well-suited for tasks requiring multi-dialectal synthesis without the need for extensive fine-tuning or speaker adaptation. Its ability to generalize across dialects while preserving pronunciation accuracy and prosody positions it as a robust, efficient, and scalable solution for Arabic TTS applications in real-world scenarios.

In conclusion, for tasks involving Arabic speech generation across dialects—with requirements for quality, adaptability, and performance—SpeechT5 offers the best balance and is strongly recommended for continued use and further experimentation.